

SYNOPSIS

CREDIT CARD FRAUD DETECTION USING MACHINE LEARNING

PROBLEM ID : 12

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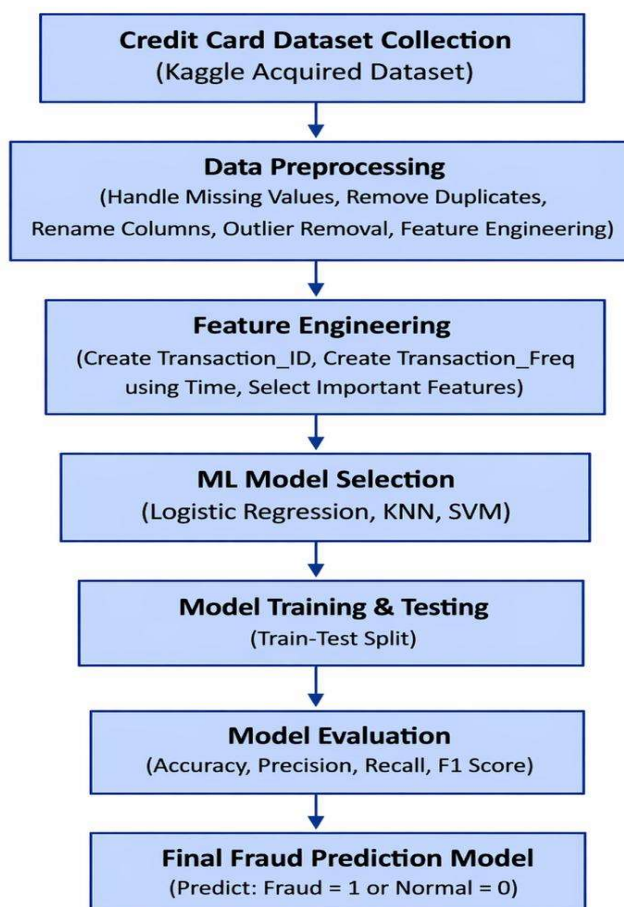
PROBLEM DESCRIPTION:

In today's digital time, most of the people use credit cards or debit cards for different purposes like online shopping, paying bills, booking tickets, etc. This makes the work easy and fast, but at the same time it also increases the chances of fraud. Fraud means when someone uses the card details without permission and makes transactions. In our dataset, there are different columns such as transaction id, amount, time, merchant type, transaction frequency and class. The class column tells whether the transaction is **legitimate [normal](0)** or **fraudulent [fraud](1)**.

In this project, we are trying to solve this problem using machine learning techniques. We will build a system which can analyse the transaction data and predict whether a transaction is fraud or not. First, we will clean the dataset by checking missing values, removing duplicates and doing normalization wherever required.

Then, we will apply different machine learning algorithms such as Logistic Regression, KNN and SVM. After training the models, we will test them and evaluate their performance using different metrics to get better and more accurate results.

STEPS OF IMPEMETATION:



STEPS OF PREPROCESSING:

1. Data Preprocessing:

a) Selecting Required Data (Row Reduction):

From the large dataset, only the first **12,000 rows** are selected using the **head()** function. This helps in reducing computational complexity and makes processing faster.

b) Column Renaming: The dataset columns are renamed to meaningful names for better understanding

- step → Time
- amount → Amount
- type → Transaction_Type
- isFraud → Class

c) Handling Missing Values:

Missing values in numerical columns are filled using the median of each column using the **fillna()** function. This helps in maintaining data consistency and avoids errors during processing.

d) Removing Duplicate Rows:

Duplicate rows are removed using the **drop_duplicates()** function to avoid repetition and ensure better data quality.

e) Outlier Detection and Removal:

Outliers in the Amount column are detected using the **Interquartile Range (IQR) method**. Values outside the range **(Q1 – 1.5 × IQR)** and **(Q3 + 1.5 × IQR)** are removed to improve data quality, model performance and reduce noise.

2. Feature Engineering:

a) Transaction ID Creation:

A new column **Transaction_ID** is created using sequential numbering to uniquely identify each transaction.

b) Transaction Frequency Creation:

A new feature **Transaction_Freq** is created based on the number of transactions occurring at the same time using grouping operation. The values are scaled down using simple division to keep them in a smaller range.

c) Selection of Relevant Features:

Only important columns are selected for the final dataset:

Transaction_ID, Time, Amount, Transaction_Type, Transaction_Freq, and Class

EXPECTED MODULES/LIBRARIES TO BE USED:

- 1. Pandas:** The Pandas library in Python is primarily used for high-performance data manipulation and analysis, utilizing Series and DataFrames to manage structured data. In this project, it is used to load the dataset, select required rows, handle missing values, remove duplicates, and perform preprocessing operations.
- 2. NumPy:** NumPy is primarily used for array processing, linear algebra, and statistical analysis, and acts as the foundational package for data science libraries like Pandas and Scikit-learn. It supports numerical computations required during data preprocessing.
- 3. Matplotlib / Seaborn:** Matplotlib is an open-source Python library used for creating visualizations like histograms and boxplots to understand data distribution and detect outliers. Seaborn is built on Matplotlib and is used for creating more advanced statistical visualizations, though basic plots are mainly used in this project.
- 4. Scikit-learn (sklearn):**
Scikit-learn is used for implementing machine learning algorithms like Logistic Regression, KNN, and SVM. It also provides tools for data splitting and evaluation metrics for model performance.
- 5. Label Encoder (sklearn.preprocessing):** Label Encoder is used to convert categorical data such as transaction types into numerical form so that machine learning models can process them.
- 6. Train test split (sklearn.model_selection):** This module is used to split the dataset into training and testing sets, which helps in validating the performance of the machine learning model.
- 7. Metrics (sklearn.metrics):** Metrics are used to evaluate model performance using measures such as accuracy, precision, recall, F1-score, and confusion matrix.
- 8. Datetime (Pandas datetime functions):** Datetime functions can be used to extract time-based features if required. In this dataset, the Time column is already in numerical format and used directly for analysis.
- 9. Custom Functions:** User-defined logic is applied for tasks such as outlier removal using the IQR method and creation of new features like transaction frequency.

EXPECTED EVALUATION METRICS AND PLOTS TO BE USED:

1.Accuracy : It is used in machine learning that measures the proportion of correct predictions (both true positives and true negatives) out of the total number of predictions made.

$$\text{Accuracy formula} = (\text{Correct predictions}) / (\text{Total predictions}).$$

2. Precision: It measures the accuracy of positive predictions, calculated as the ratio of true positives to all predicted positives

$$\text{Precision formula} = \text{TP} / (\text{TP} + \text{FP}).$$

Where:

TP = True Positive. , TN = True Negative
FP = False Positive. , FN = False Negative

3. Recall (Sensitivity): It measures the ability of the model to correctly identify all positive instances.

$$\text{Recall formula} = \text{TP} / (\text{TP} + \text{FN}).$$

4. F1-Score: It is used for calculating the harmonic mean of precision and recall.

$$\text{F1-score formula} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall}).$$

5. Confusion Matrix: A confusion matrix is a table showing actual vs predicted values (TP, TN, FP, FN). It helps visualize model performance and types of errors.

	Predicted Positive	Predicted Negative
Actual Positive	TP (True Positive)	FN (False Negative)
Actual Negative	FP (False Positive)	TN (True Negative)

PLOTS REQUIRED:

1. Class Distribution Plot

- Shows the number of fraud vs non-fraud transactions.
- Helps identify data imbalance, which is common in fraud datasets.

2. Correlation Heatmap

- Displays relationships between features using colors.
- Helps in identifying which features are strongly related and useful for fraud detection.

3. Feature Distribution (Histogram/Boxplot)

- Shows how values of features are spread.
- Used to visualize the distribution of numerical features such as transaction amount.

4. Boxplot

- Used to detect outliers in the Amount column.
- It helps in visualizing extreme values and comparing the data before and after outlier removal

REQUIRED TOPICS FROM THE SUBJECT:

▪ Python Programming

Python is used as the main programming language for implementing the machine learning pipeline such as data preprocessing, feature engineering, and preparing the dataset for fraud detection.

▪ Data Structures

Data structures like arrays and DataFrames are used to store and handle transaction data efficiently. Pandas DataFrame is mainly used for managing and manipulating the dataset.

▪ File Handling

File handling is used to read the dataset from CSV files and store the processed dataset after preprocessing using functions like **read_csv()** and **to_csv()**.

▪ Functions

Built-in functions are used for tasks like handling missing values, removing duplicates, and calculating transaction frequency. These functions help in simplifying and organizing the code.

▪ Conditional Statements

Conditional statements are used while filtering data, such as selecting fraud transactions and applying conditions during preprocessing steps like outlier removal.

- **Loops**

Loops are used for performing repetitive operations on the dataset, such as applying preprocessing steps on columns when required.

- **Machine Learning Concepts**

Concepts like classification are used to identify whether a transaction is fraudulent or not. Models like Logistic Regression, KNN, and SVM are applied for prediction.

- **Data Visualization**

Visualization techniques are used to understand data distribution and detect patterns using plots like histogram, boxplot, and correlation heatmap.

PLATFORM REQUIRED:

Jupyter Notebook: An interactive development environment used to write and execute Python code. It allows easy visualization, testing, and step-by-step implementation of machine learning models.

BOOKS AND LINK SOURCES:

1. <https://www.kaggle.com>

Used for downloading the Credit Card Fraud Detection Dataset.

2. <https://scikit-learn.org>

Official documentation for machine learning algorithms.

Hands-On Machine Learning with Scikit-Learn (reference book)

3. <https://pandas.pydata.org>

Refrence for data preprocessing using Pandas.

4. <https://www.geeksforgeeks.org>

For basic concepts and examples.

5. <https://matplotlib.org>

Used for understanding data visualization techniques.